

ODD-CYCLE COLOURINGS OF ABELIAN CAYLEY GRAPHS COME FROM CHARACTERS

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ABSTRACT. We show that a Cayley graph of an abelian group has a homomorphism to the odd cycle C_{2k+1} if and only if some character maps every generator into the closed arc $[k/(2k+1), (k+1)/(2k+1)]$ of $\mathbb{R}/\mathbb{Z}$. In particular such a graph is 3-colourable if and only if some character maps every generator into $[1/3, 2/3]$. The proof averages the height function of a colouring over an ergodic translation-invariant measure. Consequences: Payan's theorem that a cube-like graph is never 3-chromatic, and its extension by Krebs and Sankar to groups of exponent 4, follow in two lines; a distance graph $G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; and for a number field F the plane F^2 is 3-colourable if and only if a character of the group generated by the unit vectors maps them into $[1/3, 2/3]$. For $F = \mathbb{Q}(\sqrt{d})$ the last condition is a question on a 4-dimensional torus, which exact branch-and-bound certificates settle. We obtain $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$, and $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for $d = 143, 167$, five fields for which searches for finite non-3-colourable unit-distance graphs had failed.

1. INTRODUCTION

For an abelian group Γ and a subset $S = -S$ of $\Gamma \setminus \{0\}$, the Cayley graph $\text{Cay}(\Gamma, S)$ has vertex set Γ and an edge between g and $g + s$ for $g \in \Gamma, s \in S$. A *character* is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$. For $x \in \mathbb{R}/\mathbb{Z}$ let $\|x\|$ be its distance to 0.

Theorem 1. *Let $k \geq 1$ and $n = 2k + 1$. The graph $\text{Cay}(\Gamma, S)$ has a homomorphism to the cycle C_n if and only if there is a character ξ of Γ with $\xi(s) \in [k/n, (k+1)/n]$ for every $s \in S$. In particular, $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps every element of S into $[1/3, 2/3]$.*

A homomorphism to C_{2k+1} is a circular $(2k+1)/k$ -colouring, so with $\kappa(S) = \max_{\xi} \min_{s \in S} \|\xi(s)\|$ (the maximum exists, as the dual group is compact) the theorem says

$$\chi_c(\text{Cay}(\Gamma, S)) \leq 2 + \frac{1}{k} \iff \kappa(S) \geq \frac{k}{2k+1} \quad (k \geq 1).$$

The *if* direction is the classical inequality $\chi_c \leq 1/\kappa$, which for distance graphs goes back to Zhu [15] and the earlier work on distance graphs surveyed by Liu [11]. The content of the theorem is the converse at the thresholds $2 + 1/k$, among them $\chi \leq 3$. We have not found it in the literature (Section 7).

Section 2 proves the theorem. Section 3 derives Payan's theorem [13] and the exponent-4 case of Krebs and Sankar [10]. Section 4 treats distance graphs. Section 5 applies the theorem to the unit-distance graph of the plane over a number field, the setting that led to it, and Section 6 records that nothing of the kind holds for four colours.

2. PROOF OF THEOREM 1

If. Let $c(g) = 2\lfloor n\xi(g) \rfloor \bmod n$, where $\xi(g)$ is read in $[0, 1)$. Along an edge, $n\xi$ moves by an amount in $[k, k+1]$ modulo n , so $\lfloor n\xi \rfloor$ moves by k or $k+1$ modulo n , and c moves by $2k \equiv -1$ or $2k+2 \equiv 1$. So c is a homomorphism to C_n , whose vertices are $\mathbb{Z}/n$ with $j \sim j \pm 1$.

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Only if. Let $c : \Gamma \rightarrow \mathbb{Z}/n$ be a homomorphism to C_n . A character of the subgroup Γ' generated by S extends to Γ , since $\mathbb{R}/\mathbb{Z}$ is divisible, so we may replace Γ by Γ' . For $g \in \Gamma'$ and $s \in S$ let $\sigma(g, s) \in \{1, -1\}$ be the lift of $c(g + s) - c(g)$; then $\sigma(g + s, -s) = -\sigma(g, s)$.

Step 1: squares do not wind. For $s, t \in S$ the two integers $\sigma(g, s) + \sigma(g + s, t)$ and $\sigma(g, t) + \sigma(g + t, s)$ lie in $\{-2, 0, 2\}$ and are congruent to $c(g + s + t) - c(g)$ modulo n . Two distinct elements of $\{-2, 0, 2\}$ differ by 2 or 4, which n (odd, at least 3) does not divide; so they are equal.

Step 2: the winding of a closed walk. Let Z be the abelian group generated by symbols $[s]$, $s \in S$, with $[-s] = -[s]$, and let R be the kernel of $Z \rightarrow \Gamma'$, $[s] \mapsto s$. A closed walk from g with steps $s_1, \dots, s_N$ ($\sum s_j = 0$) has the sum $\sum_j \sigma(g + s_1 + \dots + s_{j-1}, s_j) \equiv 0 \pmod{n}$; write it as $n\omega$. By Step 1, ω does not change when two consecutive steps are exchanged, nor when a step s immediately followed by $-s$ is deleted. Hence ω depends only on the base point and on the class $r = \sum_j [s_j] \in R$; and conjugating by a step ($g \rightarrow g + s$, the walk, back) shows that it does not depend on the base point. So each colouring c defines $\omega_c : R \rightarrow \mathbb{Z}$, which is additive (concatenate walks). Since $n\omega$ is a sum of N odd numbers and n is odd, $\omega_c(r) \equiv N \pmod{2}$ for every walk of length N representing r ; $N \equiv \sum_s m_s \pmod{2}$ for $r = \sum_s m_s [s]$ (with s running over one element of each pair $\pm s$).

Step 3: averaging. The homomorphisms $\Gamma' \rightarrow C_n$ form a non-empty closed subset X of the compact space $(\mathbb{Z}/n)^{\Gamma'}$, invariant under the translations $c \mapsto c(\cdot + h)$. As Γ' is abelian, hence amenable, X carries a translation-invariant Borel probability measure; by the Krein–Milman theorem the convex set of such measures has an extreme point μ , and an extreme invariant measure is ergodic. For fixed r the function $c \mapsto \omega_c(r)$ is continuous (it depends on finitely many coordinates) and translation-invariant, by Step 2; so it equals a constant $\omega(r)$ μ -almost surely. Let

$$f(s) = \int_X \sigma_c(0, s) d\mu(c) \in [-1, 1].$$

By invariance $\int \sigma_c(g, s) d\mu = f(s)$ for every g , and $f(-s) = -f(s)$. Integrating the identity of Step 2 over μ gives

$$\sum_{j=1}^N f(s_j) = n\omega(r) \quad \text{for every closed walk } (s_1, \dots, s_N) \text{ representing } r \in R.$$

Step 4: the character. Put $\xi(s) = \frac{1}{2}(f(s)/n + 1) \in [k/n, (k+1)/n]$; then $\xi(-s) = 1 - \xi(s)$, so ξ is well defined modulo 1 on Z . For $r \in R$ represented by a walk of length N , $\sum_j \xi(s_j) = \frac{1}{2}(\omega(r) + N)$, an integer by the parity in Step 2. So ξ vanishes on R modulo 1 and defines a character of $Z/R = \Gamma'$, with $\xi(S) \subseteq [k/n, (k+1)/n]$. If some s has $2s = 0$, then $f(s) = f(-s) = -f(s)$ gives $\xi(s) = 1/2$, consistently. $\square$

Remark 2. For $\Gamma = \mathbb{Z}^2$ with the standard generators this is the height-function picture of 3-colourings of the square lattice, in which the height changes by ± 1 along edges and by 0 around squares; ergodic measures on height functions have a well-defined slope (compare [14]). Theorem 1 says that the slopes of an ergodic measure on colourings of any abelian Cayley graph form a character as above. In general the colouring c itself need not come from a character: for $\Gamma = \mathbb{Z}^2$ most 3-colourings do not.

3. GROUPS OF SMALL EXPONENT

Corollary 3. *Let Γ be an abelian group of exponent 2 or 4. Then no Cayley graph of Γ has chromatic number 3.*

Proof. A character of Γ takes values in $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$, and the only such value in $[1/3, 2/3]$ is $1/2$. So if $\text{Cay}(\Gamma, S)$ is 3-colourable, a character maps S to $1/2$, and $g \mapsto \lfloor 2\xi(g) \rfloor$ is a proper 2-colouring. $\square$

For exponent 2 this is Payan's theorem on cube-like graphs [13], reproved with a matrix method by Cervantes and Krebs [2]; for exponent 4 it is the hard half of the main theorem of Krebs and Sankar [10], who show that an abelian group has a 3-chromatic Cayley graph if and

only if its exponent is not 1, 2 or 4, using the fundamental group of the neighbourhood complex. Corollary 3 needs neither finite generation nor finiteness. The same argument gives, for instance, that a Cayley graph of an abelian group of exponent 8 which maps to C_5 is bipartite, since $1/2$ is the only element of $\frac{1}{8}\mathbb{Z}/\mathbb{Z}$ in $[2/5, 3/5]$; for C_3 the values $3/8$ and $5/8$ are also available, and indeed such groups have 3-chromatic Cayley graphs [10].

4. DISTANCE GRAPHS

For a finite set D of positive integers, the distance graph $G(\mathbb{Z}, D)$ is $\text{Cay}(\mathbb{Z}, \pm D)$. Its characters are $x \mapsto \alpha x$, $\alpha \in \mathbb{R}/\mathbb{Z}$, and $\kappa(D) = \max_{\alpha} \min_{d \in D} \|\alpha d\|$ is the parameter of the lonely runner problem [11].

Corollary 4. *$G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; it maps to C_{2k+1} if and only if $\kappa(D) \geq k/(2k+1)$.*

The set of α with $\alpha d \in [1/3, 2/3]$ for all d is a finite union of closed intervals, so if it is non-empty it contains an endpoint $(j + 1/3)/d$ or $(j + 2/3)/d$; this makes the test finite. We compared it, for all 1747 sets $D \subseteq [1, 24]$ with $|D| = 3$ and $\gcd(D) = 1$, with 3-colourability of the segment $[0, 12 \max D + 60]$ decided by a SAT solver: they agree, and the 74 sets that need four colours are exactly those of Zhu's classification [15] of 3-element sets with $\chi(G(\mathbb{Z}, D)) = 4$, namely $\{1, 2, 3m\}$ and $\{a, b, a+b\}$ with $a \not\equiv b \pmod{3}$ (after division by the gcd). Liu's survey asks whether $\chi_c(G(\mathbb{Z}, D)) < 1/\kappa(D)$ can happen when $|D| = 3$ [11, Problem 3]; Corollary 4 shows that below 3 the two quantities always lie between the same consecutive thresholds $2 + 1/(k+1) < \cdot \leq 2 + 1/k$.

5. PLANES OVER NUMBER FIELDS

Let F be a field of characteristic 0 with $i \notin F$, and $L = F(i)$. Identify F^2 with L by $(x, y) \mapsto x + iy$; the unit vectors are $T = \{z \in L : z\bar{z} = 1\}$, and the unit-distance graph of F^2 is $\text{Cay}(L, T)$, a disjoint union of copies of $\text{Cay}(A, T)$ with $A = \mathbb{Z}[T]$.

Corollary 5. *$\chi(F^2) \leq 3$ if and only if some character of A maps every unit vector into $[1/3, 2/3]$. If for a finite set $U \subset T$ no character of the group $\mathbb{Z}U$ maps U into $[1/3, 2/3]$, then $\chi(F^2) \geq 4$, and some finite unit-distance graph over F is not 3-colourable.*

The last assertion is the theorem of de Bruijn and Erdős [3]. Since $\mathbb{Z}U$ is a finitely generated subgroup of the $\mathbb{Q}$ -vector space L , it is free of rank at most $[L : \mathbb{Q}]$, so for a quadratic field F the test for a finite U is a question on a 4-dimensional torus. A locally constant 3-colouring at a place of F [4, 12, 6] is a colouring of a finite quotient of A , and Theorem 1 applied to that quotient gives such a character.

5.1. Real quadratic fields. For $F = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree, $\chi(F^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [9, 4, 12], $\chi(F^2) \leq 3$ if moreover $d \not\equiv 2 \pmod{3}$ [4, 12], and $\chi(F^2) \leq 4$ if $d \equiv 3 \pmod{8}$ [4, Theorem 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [12]. In [7] we found finite non-3-colourable unit-distance graphs over $\mathbb{Q}(\sqrt{d})$ for 27 values $d \equiv 11 \pmod{12}$ below 1000, but searches for such graphs failed for $d = 83, 107, 143, 167, 203$, the values that the local-global question of [6] predicted to need four colours and that remained open below 210.

Fix a denominator D and let U_D be the unit vectors $((a + b\sqrt{d})/D, (c + e\sqrt{d})/D)$, $a, b, c, e \in \mathbb{Z}$, one of each pair $\pm u$. A character with $\xi(U_D) \subseteq [1/3, 2/3]$ gives $f \in [1/3, 2/3]^{U_D}$ with $\langle r, f \rangle \in \mathbb{Z}$ for every integer relation r among the vectors of U_D . Using an LLL-reduced basis of the relations, we rule this out by branch and bound over the integers $\langle r, f \rangle$: each leaf carries a rational vector y (a Farkas certificate) such that $\langle y, \cdot \rangle$ applied to the equations fixed at that leaf cannot vanish on the box $[1/3, 2/3]^{U_D}$. A short program checks a certificate in exact integer and rational arithmetic: the vectors have length 1 ($a^2 + db^2 + c^2 + de^2 = D^2$, $ab + ce = 0$), the relations hold, every branch covers every integer value in its range, and every Farkas vector is valid. A basis of the relations, or any set of valid relations, only weakens the conditions, so infeasibility is a proof.

Theorem 6. *$\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$; $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$; and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$.*

Proof. The lower bounds are Corollary 5 with the certificates of Table 1 [5]. The upper bound 4 holds because $83 \equiv 107 \equiv 203 \equiv 3 \pmod{8}$ [4, Theorem 10]. For $d = 143$ the place above 11 is ramified in $\mathbb{Q}(\sqrt{143})$ with residue field $\mathbb{F}_{11}$, and i is not in its completion; unit vectors are integral there and reduce to the circle $x^2 + y^2 = 1$ of $\mathbb{F}_{11}^2$, whose graph is 5-colourable [7], which colours each coset of the valuation ring. $\square$

d	D	vectors	relations	nodes	leaves
83	510	54	50	6 886	4 664
107	1170	90	86	12 785	9 618
143	1740	54	50	3 861	2 710
167	1560	54	50	8 552	6 094
203	1530	78	74	1 072	732

TABLE 1. Certificates that no character of $\mathbb{Z}U_D$ maps U_D into $[1/3, 2/3]$ (vectors counted up to sign).

With the values of [7], $\chi(\mathbb{Q}(\sqrt{d})^2)$ is now known for every squarefree $d < 143$ except $d = 47$. $\mathbb{Q}(\sqrt{167})$ is the first admissible field of [6]: no place has a locally constant 4-colouring, so $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$ is also the three-colour case of the local-global question there; for $d = 203$ the result shows that no level of the place above 7 is 3-colourable, which [6] had checked only up to level 3. For $d = 83$ and $D = 510$, colouring-guided growth had stopped at 6 008 points because a 3-colouring of the ball of radius 2 extended to all 215 478 candidate points: the obstruction is global, which is why no finite graph was found. Twenty-eight of the 54 vectors already suffice for $d = 83$. For $\mathbb{Q}(\sqrt{11})$, 16 vectors with $D = 30$ suffice, in agreement with the 76-vertex graph of [7].

All five denominators are multiples of 6, as they must be: the vectors with odd D are integral above 2, where reduction gives a 2-colouring, and those with $3 \nmid D$ are integral above 3.

Question 7. Is $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$? Equivalently, by Corollary 5 and the local results above: are the locally constant colourings the only source of characters mapping all unit vectors into $[1/3, 2/3]$?

6. FOUR COLOURS

Step 1 fails for K_4 , and so does the theorem: $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ is 4-colourable, but every character of $(\mathbb{Z}/2)^2$ vanishes on some nonzero element, so none maps the three generators into $[1/4, 3/4]$. The converse inequality $\chi_c \leq 1/\kappa$ holds for all k , so a character with $\xi(S) \subseteq [1/4, 3/4]$ still gives a 4-colouring; for the 108 unit vectors of $\mathbb{Q}(\sqrt{47})$ with denominator 240, which underlie the search for a five-chromatic triangle-free unit-distance graph in [6], there is none.

7. LITERATURE

We looked for Theorem 1 in the work on 3-colourings of abelian Cayley graphs [13, 1, 2, 10, 8], in the literature on distance graphs and the lonely runner problem through Liu's survey [11], and in web searches on characters, circular colourings and odd-cycle homomorphisms of Cayley graphs. We did not find it. The proof uses only classical tools, so an earlier occurrence is possible.

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